

MONTREAL ST-HUBERT, QC, Canada

WMO: 713710

Lat: **45.520N**Lon: **73.420W**Elev: **27**StdP: **101.00**Time Zone: **-5.00 (NAE)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.2	-20.6	-29.2	0.3	-22.5	-26.6	0.3	-20.1	14.2	-2.6	12.7	-4.7	3.7	270	0.607

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	30.2	22.2	28.7	21.2	27.3	20.4	23.6	28.2	22.6	26.8	21.7	25.6	5.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.2	16.9	26.2	21.2	15.9	25.1	20.3	15.0	24.3	70.9	28.1	66.9	26.8	63.5	25.8	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.3	10.0	8.9	-26.5	32.7	3.4	1.6	-28.9	33.9	-30.9	34.8	-32.8	35.7	-35.2	36.9		
(5)			-26.8	25.6	3.3	1.2	-29.1	26.4	-31.0	27.1	-32.8	27.8	-35.2	28.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.1	-9.0	-7.4	-1.8	6.0	13.6	18.7	21.3	20.2	16.0	9.0	2.2	-4.7
	DBStd	11.65	7.03	6.01	5.71	4.60	4.21	3.68	2.84	2.98	4.07	4.41	5.00	6.13
	HDD10.0	2361	589	487	368	135	13	0	0	0	2	74	238	455
	HDD18.3	4361	847	720	626	370	156	40	6	15	93	291	485	713
	CDD10.0	1296	0	0	1	16	125	260	351	317	182	42	3	0
	CDD18.3	256	0	0	0	1	9	50	98	74	22	1	0	0
	CDH23.3	1909	0	0	1	9	107	396	726	508	159	4	0	0
	CDH26.7	441	0	0	0	1	22	101	179	107	32	0	0	0
Wind	WSAvg	4.3	4.8	4.7	4.7	4.8	4.3	3.9	3.7	3.5	3.7	4.1	4.4	4.5
Precipitation	PrecAvg	1021	77	58	65	84	90	100	102	96	89	99	77	89
	PrecMax	1266	141	103	120	152	183	195	157	179	197	183	140	166
	PrecMin	824	27	22	18	17	43	45	53	39	25	30	12	27
	PrecStd	108	32	21	25	37	33	35	26	32	37	41	30	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.8	7.2	15.6	24.2	29.2	31.6	32.3	31.2	30.1	23.7	17.0	11.0
		MCWB	7.9	5.2	10.4	16.4	19.4	22.7	23.6	23.0	22.5	18.0	14.0	9.0
	2%	DB	5.1	4.8	10.7	19.8	26.3	29.4	30.2	29.2	27.1	20.8	14.1	7.5
		MCWB	4.0	3.0	6.8	12.4	17.3	21.5	22.5	21.7	20.7	16.2	11.8	6.2
	5%	DB	3.0	3.0	7.8	16.6	23.8	27.5	28.6	27.7	25.0	18.2	11.6	5.2
		MCWB	1.9	1.5	4.9	10.4	15.9	20.1	21.5	20.8	19.5	14.2	9.3	3.8
	10%	DB	1.1	1.6	5.6	13.8	21.5	25.6	27.2	26.2	23.0	15.9	9.5	3.2
		MCWB	0.1	0.5	3.2	8.5	15.0	19.0	20.7	20.2	18.5	12.7	7.3	2.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.1	5.8	11.9	17.1	21.1	24.6	25.3	24.5	23.4	19.2	14.8	9.6
		MCDB	8.5	7.2	14.8	22.3	27.5	29.9	30.8	29.3	27.7	22.4	16.2	10.8
	2%	WB	4.2	3.2	7.4	13.8	19.2	22.8	23.8	23.1	21.8	16.7	12.0	6.4
		MCDB	4.9	4.5	10.0	18.1	23.8	27.6	28.4	27.2	25.5	19.8	14.1	7.4
	5%	WB	2.0	1.7	5.2	11.1	17.6	21.4	22.7	22.1	20.4	14.7	9.7	4.0
		MCDB	3.0	2.8	7.3	15.4	21.9	25.5	26.7	25.9	24.1	17.7	11.2	5.0
	10%	WB	0.2	0.6	3.4	9.2	15.9	20.2	21.8	21.1	19.0	13.1	7.5	2.2
		MCDB	1.1	1.6	5.3	13.2	19.9	24.2	25.7	24.7	22.4	15.5	9.2	3.1
Mean Daily Temperature Range	5% DB	MDBR	8.1	8.2	8.3	9.9	10.8	10.1	9.8	10.1	10.4	8.6	7.3	7.0
		MCDBR	9.0	8.7	11.0	15.2	14.2	12.4	11.2	11.3	12.1	11.9	10.3	8.0
		MCWBR	8.7	8.0	8.0	9.5	7.6	6.5	5.7	5.7	6.7	7.8	8.6	7.8
	5% WB	MCDBR	9.1	8.2	10.2	13.0	11.8	10.4	9.7	9.9	10.5	10.6	9.8	8.1
		MCWBR	8.8	7.7	8.1	9.1	7.5	6.4	5.8	5.7	6.6	7.7	8.8	7.9
Clear-Sky Solar Irradiance	taub	0.260	0.290	0.332	0.365	0.392	0.440	0.432	0.417	0.376	0.336	0.308	0.296	
	taud	2.350	2.252	2.289	2.349	2.330	2.203	2.230	2.268	2.370	2.486	2.481	2.297	
	Ebn at Noon	859	888	895	894	876	832	833	832	839	831	795	756	
	Edh at Noon	74	100	113	117	123	140	135	125	104	80	67	72	
All-Sky Solar Radiation	RadAvg	1.45	2.40	3.63	4.36	5.12	5.63	5.65	5.00	3.96	2.28	1.38	1.00	
	RadStd	0.16	0.25	0.29	0.51	0.48	0.42	0.42	0.23	0.33	0.20	0.18	0.09	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (43 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+61	N/A

Nomenclature: See separate page